浒苔对重金属 Pb^{2+} 的生物吸附及其生理反应

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摘要：研究了浒苔对 Pb^{2+} 的生物吸附能力，以及不同浓度 Pb^{2+} 对浒苔的叶绿素含量、叶绿素 a/b 比值、可溶性糖和蛋白质含量的影响。结果表明，暴露 72 h 后，浒苔对 Pb^{2+} 的去除率平均为（84.44 ± 0.99）%，本试验范围内呈先升高后降低的趋势。叶绿素 a、b 含量在 Pb^{2+} 不同浓度处理下呈现一定波动，但叶绿素 a/b 的比值变化不大。可溶性糖含量与空白对照相比总体上增加，而蛋白质含量总体上略有减少，但两者均有随着 Pb^{2+} 浓度不断上升先升后降的趋势。结果显示，浒苔对 Pb^{2+} 具有较好的去除率和一定的生理抗性，可以用作低浓度含铅废水的去除。

关键词：浒苔；Pb^{2+}；生物吸附；可溶性糖；蛋白质

Biosorption of Pb^{2+} in *Enteromorpha prolifera* and their Physiological Responses

Abstract: Biosorption of Pb^{2+} by *Enteromorpha prolifera* was investigated, and the chlorophyll content, chlorophyll a/b ratio, soluble sugar content and protein content of *E. prolifera* were measured at different concentrations of Pb^{2+}. Results showed that Pb^{2+} removal rate from aqueous solutions, after 72 h exposure, increased firstly and then decreased, with the average value of (84.44 ± 0.99) %. The chlorophyll contents of *E. prolifera* fluctuated at different concentrations of Pb^{2+}, whereas chlorophyll a/b ratios were constant comparatively. The soluble sugar contents of *E. prolifera* increased at different concentrations of Pb^{2+}, as compared to the control group. However, the protein contents of *E. prolifera* decreased slightly. But both increased firstly and then decreased with the increase of Pb^{2+} concentration. The experimental results suggested that *E. prolifera* had higher biosroption capacity and physiological resistance to Pb^{2+}. It can be considered as biosorbent materials for Pb^{2+} at low concentrations.

Key words: *Enteromorpha prolifera*; Pb^{2+}; biosorption; soluble sugar; protein

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